

Assignment:-1

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Subject: Software Engineering

① Explain Software Crisis and the selection of Life Cycle model.

→ The term emerged during the late 1960s when software systems became large and complex.

Root causes:-

- Rapid increase in software demand
- Lack of skilled developers
- No standard development methodologies
- Poor communication between stakeholders
- Hardware evolved faster than software practices

Selection of life cycle model totally depends on the use case of the developer there is no any fixed life cycle model, that structure can be defined accordingly to the need.

Although there are key decision factors like:

- Requirement stability
- Technical complexity
- Risk Handling capabilities
- User interaction level

② ~~Define~~ ^{Demonstrate} prototype model with its advantages & disadvantages.

→ Prototype model is based on the idea of developing an initial implementation for user feedback, and then refining the prototype through many versions until a satisfactory system emerges.

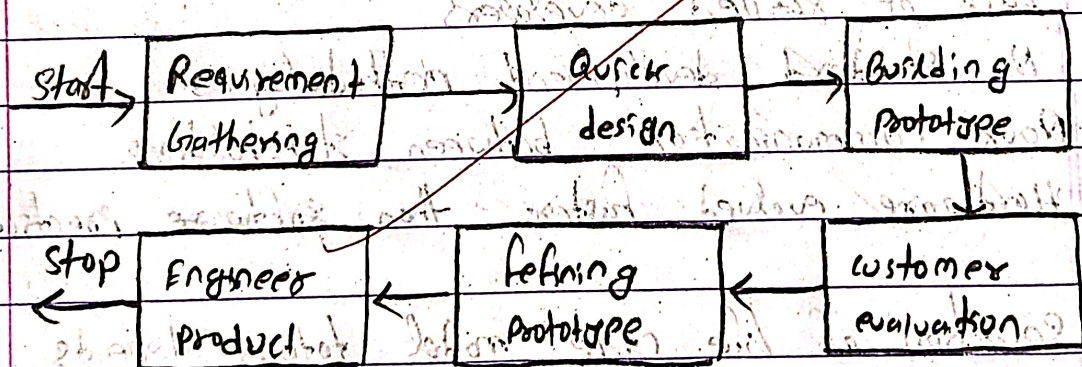


Fig: prototype model life-cycle

Advantages:

- Better understanding of requirements
- Early user involvement
- Reduces risk of failure
- Improves user satisfaction

Disadvantages

- Time-consuming
- Can increase cost
- Developers may ignore proper design
- Users may expect final system too early

② Define the advantages and disadvantages of waterfall model

→ Waterfall model emphasizes the software developed in sequential phase (e.g. Analysis, design, code etc). with established milestones, documents, and reviews at the end of each phase.

Advantages:

- Simple & easy to understand
- Well-structured phases
- Good for small & stable projects
- Documentation is clear

Disadvantages:

- No Flexibility
- Late testing
- Not suitable for changing requirements
- client feedback comes very late

④ Analyze different software process models and suggest the best model for a banking system.

→ The spiral model is a risk driven, iterative process model that combine elements of both design and prototype. Development proceeds in cycles,

- Planning
- Risk analysis
- Development
- Evaluation

It is mainly used for large, complex, and high-risk system,

Points:

- Strong risk management at every step stage.
- Continuous testing and validation.
- Suitable for secure and critical systems.
- Handles complexity effectively
- Expensive and requires expert analysis.
- Time-consuming compared to simpler models.

Agile is an iterative and incremental model where development is done in small cycles called sprints, with continuous customer feedback and rapid delivery of features.

Points:

- Highly Flexible and adaptive to changes
- Enables fast delivery of working software
- Continuous user involvement and feedback
- Improves customer satisfaction
- Less emphasis on documentation
- Weak in risk management and security control for critical systems.

Conclusion:

Spiral is best for banking:

- It focuses on risk analysis,
- Provides high security and reliability
- Suitable for large and complex applications.

Agile

- Customer-facing modules (UI)
- Features requiring frequent updates.

⑤ Discuss Formal Methods and their importance.

→ Formal methods use mathematical techniques to specify, develop, and verify software.

Semantic Static Analysis	Abstract Interpretation	Model Checking
Proof Assistants	Different Types of Formal Methods	Deductive Verification
Design by Refinement		Model-Based Testing

Types:

- Formal specification: defining system mathematically
- Formal verification: proving correctness

Importance:

- Ensure high reliability
- Detects error early
- Used in critical systems
- Provides proof of correctness

⑥ Explain 4th Generation Techniques (4GT) in Software development.

→ 4GT refers to a class of software development methods that use high-level, non-procedural language and tools to automatically generate codes.

These techniques focus on rapid development, reduced coding effort, and higher productivity.

Features of 4GT

- uses high-level languages
- Includes automated code generation tools
- Supports rapid application development (RAD)
- Requires less manual coding.

Examples of 4GT Tools

- SQL (for database queries)
- Report generators
- Application generators
- GUI-based development tools
- Low code / No code platforms.

⑦ Analyze the importance of software engineering standards.

→ Standards are rules / guidelines followed during software development

Types of standards

- Process standard :- how software is developed
- Product standards :- quality of final product

Key organizations:

- ISO (International organization for standardization)
- IEEE (Institute of Electrical and Electronics Engineers)

Importance:-

- Helps in certification (ISO 9001)
- Required for global software outsourcing
- Ensures interoperability between systems.
- Improves quality and makes maintenance easier.

